

loops

```
In [1]: print('data science')
        print('data science')
        print('data science')
        print('data science')
        print('data science')
```

data science
data science
data science
data science
data science

```
In [2]: i = 1          # initializing

        while i<=5:    # condition
            print('data science')
            i = i + 1  # increment
```

data science
data science
data science
data science
data science

```
In [3]: i = 5          # initializing

        while i>=1:    # condition
            print('data science')
            i = i - 1  # decrement
```

data science
data science
data science
data science
data science

```
In [4]: i = 1          # initializing
        while i<=5:    # condition
            print('data science : ', i)
            i = i + 1  # increment
```

data science : 1
data science : 2
data science : 3
data science : 4
data science : 5

```
In [5]: i = 5          # initializing
        while i>=1:    # condition
            print('data science : ',i)
            i = i - 1  # decrement
```

```

data science : 5
data science : 4
data science : 3
data science : 2
data science : 1

```

```

In [6]: i = 1

while i<=5:
    print('data science') # when we mention end then new line will not create
    j = 1
    while j<=4:
        print('technology')
        j = j + 1

    i = i + 1
    print()

    # the output which we got is very lengthy but how to make them one line Lets

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

In [7]: i = 1
while i<=5:
    print(' datascience', end = "") # when we mention end then new line will not
    j = 1
    while j<=4:
        print(' technology', end="")
        j = j + 1

```

```
i = i + 1
print()
```

```
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
```

In [8]: *# Lets use while loop using some numbers*

```
i = 1

while i <= 2 :
    j = 0
    while j <= 2 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
0 1 2
0 2 4
```

In [9]: *# Lets use while loop using some numbers*

```
i = 1
while i <= 4 :
    j = 0
    while j <= 3 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

In [10]: `name = 'nit'`

```
for i in name:
    print(i)
```

```
n
i
t
```

In [11]: `name1 = [1,3.5,'hallo']` *#i want print the value individually*

```
for i in name1:
    print(i)
```

```
1
3.5
hallo
```

In [12]: `for i in [2, 3, 7.8, 'hi']:`
`print(i)`

2
3
7.8
hi

In [13]: `range(5)`

Out[13]: `range(0, 5)`

In [14]: `for i in range(5):
 print(i)`

0
1
2
3
4

In [15]: `for i in range(2,5):
 print(i)`

2
3
4

In [16]: `for i in range(1,10,3):
 print(i)`

1
4
7

In [17]: *# print the value which is divisible by 5*

```
for i in range(1,51):  
    if i%5==0 :  
        print(i)
```

5
10
15
20
25
30
35
40
45
50

In [18]: *# print the value which is divisible by 5 i dont want that value*

```
for i in range(1,51):  
    if i%5!=0 :  
        print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
29
31
32
33
34
36
37
38
39
41
42
43
44
46
47
48
49

In [19]: *# write the code user ask chocklet from vendor machne write the basic code*

```
x = int(input('How many choclets you want:?'))  
  
i = 1  
while i<=x:  
    print('choclet')  
    i += 1
```

choclet
choclet
choclet
choclet
choclet

In [20]: **for** i **in** range(1,11):
 print(i)

```
1
2
3
4
5
6
7
8
9
10
```

```
In [21]: for i in range(1,11):
          print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

```
In [22]: for i in range(1,11):
          if i == 3:
              continue
          print(i)
```

```
1
2
4
5
6
7
8
9
10
```

```
In [23]: for i in range(1,11):
          if i == 6:
              continue
          print('hello :',i)
```

```
hello : 1
hello : 2
hello : 3
hello : 4
hello : 5
hello : 7
hello : 8
hello : 9
hello : 10
```

```
In [24]: for i in range(1,51):

          if i%3 == 0:

              print(i)
          print('end')
```

3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end

In [25]:

```
for i in range(1,51):  
  
    if i%3 == 0:  
        continue  
    print(i)  
print('end')
```

1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end

```
In [26]: for i in range(1,51):  
        if i%3 == 0:  
            continue  
        print(i)  
        print('end')
```

```
1  
2  
4  
5  
7  
8  
10  
11  
13  
14  
16  
17  
19  
20  
22  
23  
25  
26  
28  
29  
31  
32  
34  
35  
37  
38  
40  
41  
43  
44  
46  
47  
49  
50  
end
```

```
In [27]: for i in range(1,51):  
        if i%3 == 0 or i%5 == 0:  
            continue  
        print(i)  
        #print('end')  
        # it will skip all the value which is divisible by 3 or 5
```


1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49

```
In [28]: for i in range(1,50):  
        if i%3 == 0 and i%5 == 0:  
            continue  
        print(i)  
print('end')  
# when you apply and you wont get the value which is divisible by both 3 & 5 (15
```

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
16
17
18
19
20
21
22
23
24
25
26
27
28
29
31
32
33
34
35
36
37
38
39
40
41
42
43
44
46
47
48
49
end
```

```
In [29]: # i dont want to print the values which are even numbers that means print only o

for i in range(1,51):

    if (i%2 == 0):
        #print('even')
        continue
    else:
        print(i)
print('bye')
```

```
1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
bye
```

pattern

```
In [30]: for i in range(1,5):
          i=i+1
          print('# # # # ')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [31]: for i in range(1,5):
          if i<=5:
              print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [32]: for j in range(4):
          print('#')
```

```
#
#
#
#
```

```
In [33]: for j in range(4):
          print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [34]: for j in range(4):
         print('#', end = " ")
```

```
# # # #
```

```
In [35]: for j in range(4):
         print('#', end=" ")

         for j in range(4):
             print('#', end=" ")
```

```
# # # # # # # #
```

```
In [36]: for j in range(4):
         print('#', end=" ")

         print()

         for j in range(4):
             print('#', end=" ")
```

```
# # # #
# # # #
```

```
In [37]: for i in range(4):
         for j in range(4):
             print('#', end=" ")
             print()
         # please use debug mode in pycharm
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [38]: for i in range(4):
         for j in range(i+1):
             print('#', end = " ")
             print()
```

```
#
# #
# # #
# # # #
```

```
In [39]: for i in range(1,5):
         print("# " * i)
```

```
#
# #
# # #
# # # #
```

```
In [40]: for i in range(1,5):
         for j in range(4):
             if i > j:
```

```
print("#",end=" ")
print()
```

```
#
# #
# # #
# # # #
```

In [41]: `list(range(5))`

Out[41]: `[0, 1, 2, 3, 4]`

In [42]: `for i in range(4):`
 `for j in range(i):`
 `print('#', end=" ")`
 `print()`

```
#
# #
# # #
```

In [43]: `for i in range(4):`
 `for j in range(i+1):`
 `print('#', end=" ")`
 `print()`

```
#
# #
# # #
# # # #
```

In [44]: `for i in range(4):`
 `for j in range(4-i):`
 `print('#', end=" ")`
 `print()`

```
# # # #
# # #
# #
#
```

In [45]: `for i in range(1,5):`
 `print("# "*(5-i))`

```
# # # #
# # #
# #
#
```

for else

In [46]: `nums = [12,15,18,21,26]`

```
for num in nums:
    if num % 5 == 0:
        print(num)
```

15

```
In [47]: nums = [12,14,18,21,25,20]

for num in nums:
    if num % 5 == 0:
        print(num)
```

25
20

```
In [48]: nums = [12,14,18,21,20,25]

for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

20

```
In [49]: nums = [12,14,18,21,20,25]

for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

20

```
In [50]: nums = [10,14,18,21,5,10]

for num in nums:
    if num % 5 == 0:
        print(num)
        break #it will print only 1 number then it break
```

10

```
In [52]: nums = [7,14,18,21,23,27] #hear there is no number which is divisible by 5 we go

for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

```
In [53]: nums = [7,14,18,21,23,27] #hear there is no number which is divisible by 5 we go

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found') #every iteration it cheking condition
```

Number Not Found
Number Not Found
Number Not Found
Number Not Found
Number Not Found
Number Not Found

```
In [54]: nums = [7,14] #hear there is no number which is divisible by 5 we got output as
for num in nums:
```

```

if num % 5 == 0:
    print(num)
    break
else:
    print('Number Not Found') #every iteration it cheking condition

```

Number Not Found

Number Not Found

In [55]: `nums = [7,14,18,21,23,27]` #hear there is no number which is divisible by 5 we go

```

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found') # hear else we dont write in if block but we c

```

Number Not Found

In [56]: `nums = [10,14,18,21,20,27]` #hear there is no number which is divisible by 5 we g

```

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Not Found')

```

10

In [57]: `nums = [10,14,18,21,20,27,30]` #hear there is no number which is divisible by 5 w

```

for num in nums:
    if num % 5 == 0:
        print(num)
        #break
    else:
        print('Not Found')

```

10

20

30

Not Found

In [58]: `nums = [10,14,18,21,20,27]` #hear there is no number which is divisible by 5 we g

```

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Not Found')

```

10

prime number

In [59]: `num = 13`

```

for i in range(2,num):

```

```
if num % i == 0:  
    print('Not prime Number')  
    break  
else:  
    print('Prime Number')
```

Prime Number

In []: